week5workshop taskb

Shallow Morphological Parsing (RT)

We want to include only data from `L2` `critical` trials.

Libraries

```
library(tidyverse)
```

```
Warning: package 'tidyverse' was built under R version 4.5.3
```

```
Warning: package 'ggplot2' was built under R version 4.5.3
```

```
Warning: package 'tidyr' was built under R version 4.5.3
```

```
Warning: package 'readr' was built under R version 4.5.3
```

```
Warning: package 'purrr' was built under R version 4.5.3
```

```
Warning: package 'forcats' was built under R version 4.5.3
```

```
Warning: package 'lubridate' was built under R version 4.5.3
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.0     v readr     2.2.0
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.2     v tibble    3.3.1
v lubridate 1.9.5     v tidyr     1.3.2
v purrr     1.2.1
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becor
```

```
library(ggplot2)
library(dplyr)
library(brms)
```

```
Warning: package 'brms' was built under R version 4.5.3
```

```
Loading required package: Rcpp
```

```
Warning: package 'Rcpp' was built under R version 4.5.3
```

```
Loading 'brms' package (version 2.23.0). Useful instructions
can be found by typing help('brms'). A more detailed introduction
to the package is available through vignette('brms_overview').

Attaching package: 'brms'

The following object is masked from 'package:stats':

    ar
```

Summary measures:

First group by the columns we want then we can filter them.

```
shallow <- read_csv("./data/song2020/shallow.csv")
```

```
Rows: 6500 Columns: 11
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (8): Group, ID, List, Target, Critical_Filler, Word_Nonword, Relation_ty...
dbl (3): ACC, RT, logRT

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
#tibble

shallow |>
  group_by(Group, Critical_Filler) |>
  summarise(
    mean = mean(RT, na.rm = TRUE),
    sd = sd(RT, na.rm = TRUE)
  ) |>
  filter(
    Group == "L2",
    Critical_Filler == "Critical"
  )
```

```
`summarise()` has regrouped the output.
i Summaries were computed grouped by Group and Critical_Filler.
i Output is grouped by Group.
i Use `summarise(.groups = "drop_last")` to silence this message.
i Use `summarise(.by = c(Group, Critical_Filler))` for per-operation grouping
  (`?dplyr::dplyr_by`) instead.

# A tibble: 1 x 4
# Groups:   Group [1]
  Group Critical_Filler  mean    sd
  <chr> <chr>           <dbl> <dbl>
1 L2    Critical         968.  354.
```

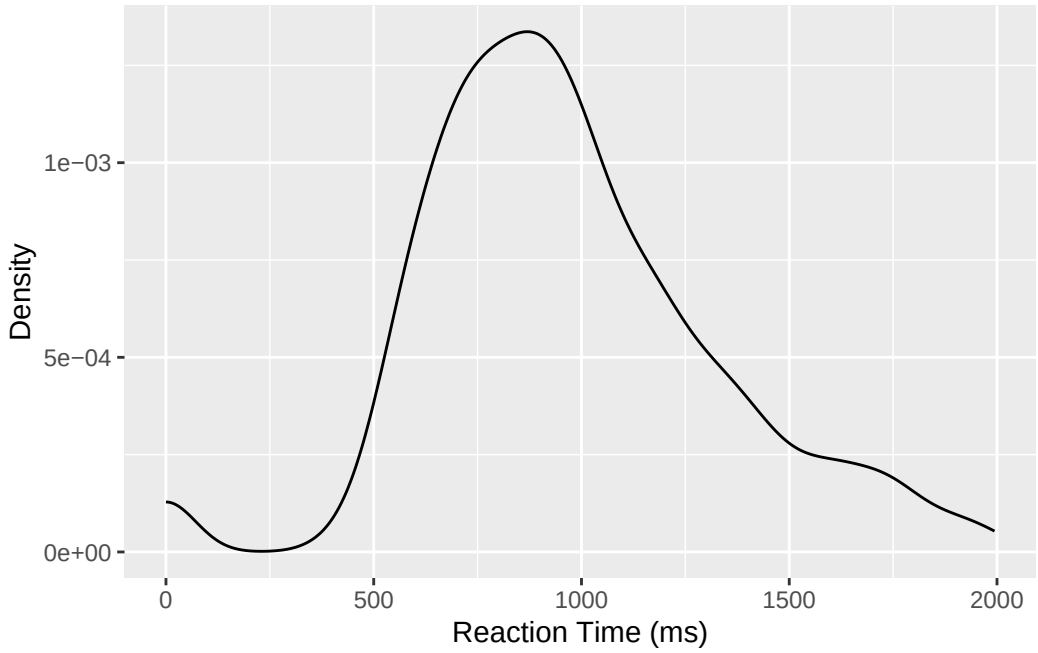
```
#df

shallow_l2_critical <- shallow |>
  filter(
    Group == "L2",
    Critical_Filler == "Critical"
  )
```

The mean reaction time of critical L2 participants is 967.75ms and the standard deviation is approximately 354.06ms.

Plot

```
shallow |>
  filter(
    Group == "L2",
    Critical_Filler == "Critical"
  ) |>
  ggplot(
    aes(x = RT)) +
  geom_density() +
  labs(
    x = "Reaction Time (ms)",
    y = "Density"
  )
```



Model

We create the model with the mean and sd values we calculated:

```r
rt_brm <- brm(
  formula = RT ~ 1,
  family = gaussian,
  data = shallow_l2_critical
)
```

```
Compiling Stan program...

WARNING: Rtools is required to build R packages, but is not currently installed.

Please download and install the appropriate version of Rtools for 4.5.2 from
https://cran.r-project.org/bin/windows/Rtools/.

Trying to compile a simple C file

Running "C:/PROGRA~1/R/R-4.5.2/bin/x64/Rcmd.exe" SHLIB foo.c
using C compiler: 'gcc.exe (GCC) 14.3.0'
gcc  -I"C:/PROGRA~1/R/R-4.5.2/include" -DNDEBUG   -I"C:/Users/lachl/AppData/Local/R/win-
library/4.5/Rcpp/include/"  -I"C:/Users/lachl/AppData/Local/R/win-library/4.5/RcppEigen/inclu
I"C:/Users/lachl/AppData/Local/R/win-library/4.5/RcppEigen/include/unsupported"  -
I"C:/Users/lachl/AppData/Local/R/win-library/4.5/BH/include" -I"C:/Users/lachl/AppData/Local/
library/4.5/StanHeaders/include/src/"  -I"C:/Users/lachl/AppData/Local/R/win-
library/4.5/StanHeaders/include/"  -I"C:/Users/lachl/AppData/Local/R/win-
library/4.5/RcppParallel/include/" -DRCPP_PARALLEL_USE_TBB=1 -DTBB_INTERFACE_NEW -
IC:/Users/lachl/AppData/Local/R/win-library/4.5/RcppParallel/include -I"C:/Users/lachl/AppDat
library/4.5/rstan/include" -DEIGEN_NO_DEBUG  -DBOOST_DISABLE_ASSERTS  -
DBOOST_PENDING_INTEGER_LOG2_HPP  -DSTAN_THREADS  -DUSE_STANC3 -DNEW_RSTAN -
DSTRICT_R_HEADERS  -DBOOST_PHOENIX_NO_VARIADIC_EXPRESSION  -D_HAS_AUTO_PTR_ETC=0  -
include "C:/Users/lachl/AppData/Local/R/win-library/4.5/StanHeaders/include/stan/math/prim/fu
D_REENTRANT -DRCPP_PARALLEL_USE_TBB=1    -I"F:/rtools45/x86_64-w64-mingw32.static.posix/inclu
O2 -Wall -std=gnu2x  -mfpmath=sse -msse2 -mstackrealign   -c foo.c -o foo.o
In file included from <command-line>:
C:/Users/lachl/AppData/Local/R/win-library/4.5/StanHeaders/include/stan/math/prim/fun/Eigen.h
    3 | #include <stdexcept>
      |          ^~~~~~~~~~~
compilation terminated.
make: *** [C:/PROGRA~1/R/R-4.5.2/etc/x64/Makeconf:289: foo.o] Error 1

WARNING: Rtools is required to build R packages, but is not currently installed.

Please download and install the appropriate version of Rtools for 4.5.2 from
https://cran.r-project.org/bin/windows/Rtools/.
```

```
Start sampling


SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 1).
Chain 1:
Chain 1: Gradient evaluation took 0.000127 seconds
Chain 1: 1000 transitions using 10 leapfrog steps per transition would take 1.27 seconds.
Chain 1: Adjust your expectations accordingly!
Chain 1:
Chain 1:
Chain 1: Iteration:    1 / 2000 [  0%]  (Warmup)
Chain 1: Iteration:  200 / 2000 [ 10%]  (Warmup)
Chain 1: Iteration:  400 / 2000 [ 20%]  (Warmup)
Chain 1: Iteration:  600 / 2000 [ 30%]  (Warmup)
Chain 1: Iteration:  800 / 2000 [ 40%]  (Warmup)
Chain 1: Iteration: 1000 / 2000 [ 50%]  (Warmup)
Chain 1: Iteration: 1001 / 2000 [ 50%]  (Sampling)
Chain 1: Iteration: 1200 / 2000 [ 60%]  (Sampling)
Chain 1: Iteration: 1400 / 2000 [ 70%]  (Sampling)
Chain 1: Iteration: 1600 / 2000 [ 80%]  (Sampling)
Chain 1: Iteration: 1800 / 2000 [ 90%]  (Sampling)
Chain 1: Iteration: 2000 / 2000 [100%]  (Sampling)
Chain 1:
Chain 1:  Elapsed Time: 0.211 seconds (Warm-up)
Chain 1:                0.086 seconds (Sampling)
Chain 1:                0.297 seconds (Total)
Chain 1:

SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 2).
Chain 2:
Chain 2: Gradient evaluation took 7.4e-05 seconds
Chain 2: 1000 transitions using 10 leapfrog steps per transition would take 0.74 seconds.
Chain 2: Adjust your expectations accordingly!
Chain 2:
Chain 2:
Chain 2: Iteration:    1 / 2000 [  0%]  (Warmup)
Chain 2: Iteration:  200 / 2000 [ 10%]  (Warmup)
Chain 2: Iteration:  400 / 2000 [ 20%]  (Warmup)
Chain 2: Iteration:  600 / 2000 [ 30%]  (Warmup)
Chain 2: Iteration:  800 / 2000 [ 40%]  (Warmup)
Chain 2: Iteration: 1000 / 2000 [ 50%]  (Warmup)
Chain 2: Iteration: 1001 / 2000 [ 50%]  (Sampling)
```

```
Chain 2: Iteration: 1200 / 2000 [ 60%]  (Sampling)
Chain 2: Iteration: 1400 / 2000 [ 70%]  (Sampling)
Chain 2: Iteration: 1600 / 2000 [ 80%]  (Sampling)
Chain 2: Iteration: 1800 / 2000 [ 90%]  (Sampling)
Chain 2: Iteration: 2000 / 2000 [100%]  (Sampling)
Chain 2:
Chain 2:  Elapsed Time: 0.186 seconds (Warm-up)
Chain 2:                0.076 seconds (Sampling)
Chain 2:                0.262 seconds (Total)
Chain 2:

SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 3).
Chain 3:
Chain 3: Gradient evaluation took 2.3e-05 seconds
Chain 3: 1000 transitions using 10 leapfrog steps per transition would take 0.23 seconds.
Chain 3: Adjust your expectations accordingly!
Chain 3:
Chain 3:
Chain 3: Iteration:    1 / 2000 [  0%]  (Warmup)
Chain 3: Iteration:  200 / 2000 [ 10%]  (Warmup)
Chain 3: Iteration:  400 / 2000 [ 20%]  (Warmup)
Chain 3: Iteration:  600 / 2000 [ 30%]  (Warmup)
Chain 3: Iteration:  800 / 2000 [ 40%]  (Warmup)
Chain 3: Iteration: 1000 / 2000 [ 50%]  (Warmup)
Chain 3: Iteration: 1001 / 2000 [ 50%]  (Sampling)
Chain 3: Iteration: 1200 / 2000 [ 60%]  (Sampling)
Chain 3: Iteration: 1400 / 2000 [ 70%]  (Sampling)
Chain 3: Iteration: 1600 / 2000 [ 80%]  (Sampling)
Chain 3: Iteration: 1800 / 2000 [ 90%]  (Sampling)
Chain 3: Iteration: 2000 / 2000 [100%]  (Sampling)
Chain 3:
Chain 3:  Elapsed Time: 0.194 seconds (Warm-up)
Chain 3:                0.085 seconds (Sampling)
Chain 3:                0.279 seconds (Total)
Chain 3:

SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 4).
Chain 4:
Chain 4: Gradient evaluation took 2.3e-05 seconds
Chain 4: 1000 transitions using 10 leapfrog steps per transition would take 0.23 seconds.
Chain 4: Adjust your expectations accordingly!
Chain 4:
Chain 4:
```

```
Chain 4: Iteration:    1 / 2000 [  0%]  (Warmup)
Chain 4: Iteration:  200 / 2000 [ 10%]  (Warmup)
Chain 4: Iteration:  400 / 2000 [ 20%]  (Warmup)
Chain 4: Iteration:  600 / 2000 [ 30%]  (Warmup)
Chain 4: Iteration:  800 / 2000 [ 40%]  (Warmup)
Chain 4: Iteration: 1000 / 2000 [ 50%]  (Warmup)
Chain 4: Iteration: 1001 / 2000 [ 50%]  (Sampling)
Chain 4: Iteration: 1200 / 2000 [ 60%]  (Sampling)
Chain 4: Iteration: 1400 / 2000 [ 70%]  (Sampling)
Chain 4: Iteration: 1600 / 2000 [ 80%]  (Sampling)
Chain 4: Iteration: 1800 / 2000 [ 90%]  (Sampling)
Chain 4: Iteration: 2000 / 2000 [100%]  (Sampling)
Chain 4:
Chain 4:  Elapsed Time: 0.176 seconds (Warm-up)
Chain 4:                0.064 seconds (Sampling)
Chain 4:                0.24 seconds (Total)
Chain 4:
```

```
summary(rt_brm)
```

```
 Family: gaussian
  Links: mu = identity
Formula: RT ~ 1
   Data: shallow_l2_critical (Number of observations: 1080)
  Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
         total post-warmup draws = 4000

Regression Coefficients:
          Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
Intercept   967.80     10.71   946.83   988.87 1.00     3733     2586

Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sigma   354.32      7.62   340.17   369.19 1.00     3819     2875

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

quartiles

```
quartiles <- qnorm(
  c(0.25, 0.5, 0.75), 967.72, 354.30)

quartiles[3] - quartiles[1]
```

```
[1] 477.9434
```

The IQR is 477.9434. This means that 50% of the density mass lies within approximately 729ms and 1207ms.

```
plot(rt_brm, combo = c("dens", "trace"))
```

